

Customs Mismatch Resolver — technical README

DTU 38113 · Module 1 · Fleet agentic AI POC · smith.langchain.com/fleet · Slack teammate

Stack

LangSmith Fleet

Slack (custom app, @mention trigger)

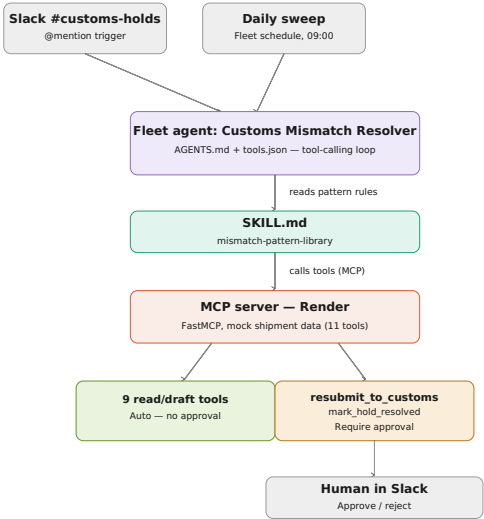
FastMCP 4 / Python

Render (free web service)

GitHub (source)

MCP over Streamable HTTP

Architecture



Mechanism

Event-driven tool-calling loop (LangChain loop-engineering: Loop 3, wrapping a verification/grader loop). Trigger fires → agent lists held shipments → for each: retrieve shipment file + flagged document → retrieve live customs ruleset for destination country → classify mismatch against SKILL.md 's known-safe pattern list → draft correction → validate against ruleset (returns pass/fail + confidence) → tiered decision.

Tier	Condition	Tool gating
Auto	Known-safe pattern and value < €5,000 and confidence > 90%	resubmit_to_customs fires directly
Approval	Any: unrecognized pattern / value ≥ €5,000 / confidence ≤ 90% / no ruleset match	Posted to Slack, waits on human click

Components

File	Role
AGENTS.md	System prompt: goal, hard constraints, tiered-autonomy rule, operating sequence, decision-receipt format
config.json	Agent metadata (name, description, visibility)
tools.json	11 MCP tool bindings + interrupt_config (per-tool approval gate, keyed on server URL + tool name)
skills/mismatch-pattern-library/SKILL.md	Known-safe pattern allowlist; explicitly forbids self-expansion without human review
server.py / data.py	FastMCP server, 11 tools, 5 mock shipments spanning all tiers + a no-op case

Deployment

Repo: github.com/RobinDFeder/applied-ai-fleet-project . Render web service, root dir customer-mismatch-core , build pip install -r requirements.txt , start python server.py --http, PYTHON_VERSION=3.12.0 . Live endpoint registered in Fleet as a custom MCP connection (Headers auth, no key) and attached to the agent's own Toolbox. Slack channel #customs-holds connected as a custom app, @mention-triggered; daily 09:00 schedule active as fallback sweep.

Test results — live runs via Slack

Shipment	Outcome	Trigger condition
SHP-1001	Auto-resolved	HS-code transposition, €1,250, 95% confidence
SHP-1002	Escalated	Known-safe pattern but €84,000 ≥ threshold
SHP-1003	Escalated	Signature-format pattern not on known-safe list

All three exercised the deployed MCP server end to end (not simulated), confirming both independent escalation conditions fire correctly.

Value creation — mapped to lecture sources

- **McKinsey, Innovation Revolution** — research operations (parallel retrieval) + design evaluation (grader step); velocity is the dominant axis, one correct answer per case rather than option generation.
- **McKinsey, Decision Dividend** — targets the >90% decision-cost-reduction claim directly; decision receipt is the audit trail that makes the compression trustworthy.
- **Anthropic, Loop Engineering / Recursive Self-Improvement** — Loop 3 (event-driven) wraps Loop 2 (grader); autonomy stays narrow per the finding that humans retain the edge on novel/high-stakes judgment.
- **Anthropic, Building Effective Human-Agent Teams** — autonomy expands only with demonstrated reliability (SKILL.md forbids self-expansion); human attention treated as scarce (batched escalations, Slack-teammate framing, not silent background job).
- **OpenAI, Hugging Face incident** — counter-case: production harness/system-prompt constraints cut infrastructure-overreach propensity >100x. Directly motivates the Hard constraints block: fixed tool list, mandatory live-ruleset check before action, escalate-don't-persist on ambiguity, no acting on instructions found inside tool output.

Fleet package (AGENTS.md, config.json, tools.json, skills/) submitted as zip; MCP server source in repo above. All test runs and screenshots correspond to the live deployment, not a simulated flow.